

Sparse Autoencoder - High-Level Project Steps

Flowchart summary of the notebook from session data to hidden behaviour segmentation

1. Load website session data

Load the session-level behavioural dataset and keep the purchase outcome only for later business evaluation.



2. Select behavioural features

Separate model input features from labels. The autoencoder learns session behaviour, not the Revenue / purchase column.



3. Preprocess features

Scale numeric features and encode categorical features so all model inputs are numeric and comparable.



4. Train baseline autoencoder

Train a normal dense autoencoder to learn reconstruction of website session behaviour.



5. Train sparse autoencoder

Add sparsity pressure in the bottleneck so fewer hidden units activate strongly for each session.



6. Compare reconstruction and sparsity

Compare validation loss, test reconstruction error, average bottleneck activation, and near-zero activation percentage.



7. Extract sparse bottleneck space

Use the sparse encoder to create compressed hidden behaviour features for each website session.



8. Visualise hidden behaviour space

Use PCA to reduce bottleneck features to two dimensions and inspect behavioural separation.



9. Segment sessions

Cluster hidden bottleneck features into behaviour segments and attach segment labels to sessions.



10. Interpret segment value

Compare segment size, purchase rate, and reconstruction error to identify useful business segments.

Final business outcome

The sparse autoencoder creates compact hidden behaviour features. These features are used to group website sessions into interpretable segments and compare purchase behaviour without training directly on the purchase label.